Multimedia Signal Processing - 2nd Module

July, 5 2017

1. **Question (8 pts)**
   Consider two discrete memoryless sources X and Y that emit symbols from the alphabets $\{x_1, x_2, x_3, x_4\}$ and $\{y_1, y_2\}$, respectively. The joint p.m.f. is

	y_1	y_2
x_1	0.15	0.05
x_2	0.1	0.3
x_3	0.1	0.2
x_4	0.05	0.05

   (a) Compute the probability mass functions of symbols of X and Y, separately.
   (b) Design the Huffman code to encode single symbols of Y, and comment the result.
   (c) Design the Huffman code to encode single symbols of X.
   (d) Considering this last scenario (i.e., X), compute: (i) the average code length of the obtained Huffman code; (ii) the average code length of the Shannon code, and; (iii) the theoretical bound.
   (e) Design the Huffman code that jointly encodes one symbol of X and one symbol of Y.
   (f) Compute again: (i) the average code length of the obtained Huffman code; (ii) the average code length of the Shannon code, and; (iii) the theoretical bound.
   (g) Compare and comment the results obtained in point (d) and point (f).
   (h) Compute mutual information $I(X;Y)$.

2. **Question (5 pts)**
   (a) Describe how speech signals can be broadly categorized, providing some examples and sketching salient spectral / temporal characteristics.
   (b) Illustrate how source-system model can be used to exploit these speech characteristics and build a vocoder. Highlight how speech is represented in the encoded domain (i.e., which is the input of the decoder).

3. **Question (5 pts)**
   (a) Introduce the concept of scalar quantization, providing the definition of scalar quantizer and showing the difference between a midtread and a midrise one.
   (b) Describe all steps of Lloyd-Max quantization process, discussing in which situations it is useful, and in which it is not.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**
   (a) Consider a hybrid DCT/DPCM video coding scheme.
   - [T] [F] The encoder can compute motion vectors using any search algorithm.
   - [T] [F] Frames can be always decoded in random order.
   - [T] [F] Motion vectors can be computed with sub-pixel accuracy.
   - [T] [F] A part from the firsts frame (intra-coded), all the other frames are typically inter-coded using prediction methods.

   (b) Consider a perceptual audio coding scheme.
   - [T] [F] Tones are only masked by the presence of other tones
   - [T] [F] Components in a critical band can affect other components in adjacent critical bands.
   - [T] [F] The absolute threshold of hearing has a minimum around 3 or 4 kHz.

   (c) Consider two discrete memoryless sources X and Y emitting sequences of samples from a finite alphabet.
   - [T] [F] Mutual information $I(X;Y)$ is positive by definition.
   - [T] [F] Entropy $H(X)$ can be greater than 1 if X is a binary source.
   - [T] [F] It is possible to lossy encode symbols of X using less than $H(X)$ bits per symbol.

5. **Question (Matlab - 10pts)**
   Consider the image file `lena.png`.
   (a) Load the 8 bit grayscale image `I` and display it on screen.
   (b) Consider the image as a single sequence of pixels. Display the probability mass function (pmf) of the pixels and compute the entropy. Is it compatible with the image being stored using 8 bit per pixel?
   (c) Consider the image as a series of distinct blocks of two adjacent pixels. Display the 2D probability mass function (pmf) and compute the entropy. Compare with the previously obtained result and comment.
   (d) Encode `I` using block-wise transform coding based on 8×8 DCT and draw rate-distortion curve. Specifically, consider:
   - scalar uniform quantization with $Q \in \{8, 12, 14, 16, 20\}$ (hint: use `round(.)` and ensure a reproduction level equal to zero).
   - PSNR as distortion measure.
   - Entropy as rate measure.

   (e) Rapidly comment the rate-distortion curve obtained (hint: what happens for $Q = 14$?).
   (f) Considering $Q = 14$, compute and display image compression residual (i.e., difference between original image and decoded one). Comment what you see.